- Applications should expose health endpoints to check their status.
- Spring Boot Actuator exposes an overall health endpoint showing the status of the application and all the components or services it might use. It also provides specialized endpoints to be used as liveness and readiness probes by Kubernetes.
- When the liveness probe is down, it means the application has entered an unrecoverable faulty state, so Kubernetes will try to restart it.
- When the readiness probe is down, the application is not ready to handle requests, so Kubernetes will stop any traffic directed to that instance.
- Metrics are numeric data about the application, measured at regular time intervals.
- Spring Boot Actuator leverages the Micrometer façade to instrument the Java code, generate metrics, and expose them through a dedicated endpoint.
- When the Prometheus client is on the classpath, Spring Boot can expose metrics in the Prometheus or OpenMetrics format.
- Using the Grafana observability stack, Prometheus aggregates and stores metrics from all applications. Then you can use Grafana to query metrics, design dashboards, and set alerts.
- Distributed tracing, a technique for tracking requests as they flow through a distributed system, lets us localize where errors occur in a distributed system and troubleshoot performance issues.
- Traces are characterized by a trace ID and are composed of multiple spans, representing steps in a transaction.